Roll Number												

Narula Institute of Technology

An Autonomous Institute under MAKAUT

2024

END SEMESTER EXAMINATION - ODD 2024

M(IT)101 - ENGINEERING MATHEMATICS - I

TIME ALLOTTED: 3Hours **FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)
Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) For what the value of t for which the matrix $\begin{bmatrix} 2 & 0 & 1 \\ 5 & t & 3 \\ 0 & 3 & 1 \end{bmatrix}$ is singular. (1) CO4 BL5

a) **0**
b) **1**
c) **2**
d) **3/2**

1.ii) If A be a square matrix of order 2. Given that Det(A) = 6 and tr(A)=5, then the eigenvalues of A (1) CO2 BL2

a) **2,2**
b) **1,2**
c) **2,3**
d) **None of these**

1.iii) The matrix $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ is (1) CO1 BL1

a) **Symmetric**
b) **Skew symmetric**
c) **Orthogonal**
d) **None of these**

1.iv) Diagonal elements of skew symmetric matrix are (1) CO1 BL1
a) **All non zero**
b) **All zero**
c) **1**
d) **None of these**

1.v) If $u(x,y) = \sin^{-1}\frac{x}{y} + \cos^{-1}\frac{x}{y}$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} =$ (1) CO1 BL1

a) **u**
b) **2u**
c) **0**
d) None of these

1.vi) $\int_0^{\pi/2}\int_0^{\pi} \sin(x+y)\,dx\,dy$ is (1) CO3 BL3

a) **1**
b) **2**
c) **21**
d) **None of these**

1.vii) $\int_2^{3}\int_4^{6} dx\,dy =$ (1) CO3 BL3

a) **0**
b) **1**
c) **2**
d) **3**

1.viii) $u(x,y)=\dfrac{\sqrt{x^2}+\sqrt{y^2}}{\sqrt{x}+\sqrt{y}}$ is a homogeneous function of degree (1) CO1 BL1

a) **1**
b) **1/2**
c) **3/2**
d) **None of these**

1.ix) If $u+v=x, uv=y$ then $\dfrac{\partial(x,y)}{\partial(u,v)} =$ (1) CO4 BL4

a)
uv

b)
$u-v$

c)
$u+v$

d) None of these

1.x) The sequence $\{(-1)^n\}$ is (1) CO1 BL1

a) **Convergent**
b) **Divergent**
c) **Oscillatory**
d) **None of these**

1.xi) $\int_C y\,dx = ?$, Where Cis the line segment joining $(0,0)$ and $(1,1)$ (1) CO4 BL4

a) **1**
b)
$\dfrac{1}{2}$

c) 2
d) **Non of these**

1.xii) If $f(x,y)$ is a homogeneous function of degree 3, then $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=?$ (1) CO1 BL1

a) 3u
b) 6u

c) u
d) None of these

Group B
(Short Answer Type Questions)
(Answer any three of the following) 3x5=15

2. If $A=\begin{pmatrix}1&2&2\\2&1&2\\2&2&1\end{pmatrix}$, then show that $A^2-4A-5I_3=O_{3\times3}$. Hence obtain A^{-1}. (5) CO2 BL2

3. Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{\pi}\sin(x+y)dxdy$ (5) CO1 BL1

4. Reduce the matrix A= $\begin{bmatrix}1&2&-2&3\\2&5&-4&6\\-1&-3&2&-2\\2&4&-1&6\end{bmatrix}$ to an equivalent Echelon matrix and hence find its rank. (5) CO3 BL3

5. Find the maximum and minimum values of the function $f(x,y)=x^3+y^3-3axy$ (5) CO2 BL2

6. Investigate the value of λ and μ, for which the system of equations (5) CO3 BL2

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

has (i) Unique solution (ii) many solutions (ii) no solution at all.

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

7a. Find for those values of k and m, the system has i) unique solution ii) no solution iii) many solutions: $x+4y+2z=1;\ 2x+7y+5z=2k;\ 4x+my+10z=2k+1$ (8) CO2 BL2

7b. Find the eigenvalues and corresponding eigenvectors of the matrix $A=\begin{pmatrix}2&1&1\\1&2&1\\0&0&1\end{pmatrix}$ (7) CO4 BL4

8a. If $u=\tan^{-1}\dfrac{x^3+y^3}{x-y}$, find the value of $x^2\dfrac{\partial^2u}{\partial x^2}+2xy\dfrac{\partial^2 f}{\partial x\partial y}+y^2\dfrac{\partial^2u}{\partial y^2}$. (5) CO3 BL3

8b. If $f(x,y)=\dfrac{xy(x^2-y^2)}{x^2+y^2},\ (x,y)\neq(0,0)$ (5) CO1 BL1

$=0,\qquad (x,y)=(0,0)$

Find from definition $f_{xy}(0,0)$ and $f_{yx}(0,0)$. Verify whether $f_{xy}(0,0)=f_{yx}(0,0)$

8c. Find the extreme points of the function $f(x,y)=x^3+y^3-3x-12y+20$ and check the nature of those extreme points. (5) CO2 BL2

9. Answer the following (15)

a) Verify Cayley-Hamilton theorem for the matrix (5) CO3 BL3

$\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$. Hence evaluate A^{-1}.

b) Examine the Consistency of the system and solve, if possible. (5) CO3 BL3

$x+y+z=1$
$2x+y+2z=2$
$3x+2y+3z=5$

c) Find the volume of revolution generated by the region enclosed by $y=\sqrt{x}$ and the line y = 1 , x =4 about x-axis. (5) CO3 BL3

10. Answer the following (15)

a) If λ is an eigen value of a real orthogonal matrix A, then prove that $\frac{1}{\lambda}$ is also an eigen value of A. (5) CO2 BL2

b) Apply Maclaurin's theorem to the function $f(x)=(1+x)^4$ to deduce that $(1+x)^4=1+4x+6x^2+4x^3+x^4$ (5) CO3 BL3

c) If $f(x,y)=x^2\tan^{-1}\left(\frac{y}{x}\right)-y^2\tan^{-1}\left(\frac{x}{y}\right)$, verify that $f_{xy}=f_{yx}$ (5) CO3 BL3

11. Answer the following (15)

a) Use mean value theorem to prove that $\frac{x}{1+x^2}<\tan^{-1}x<x \ \ when \ 0<x<\frac{\pi}{2}$ (5) CO3 BL3

b) Reduce the matrix A= $\begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$ to an equivalent Echelon matrix and hence find its rank. (5) CO4 BL4

c) If f(x, y)=$\frac{x+y}{1-xy}$ and g(x,y)=$\tan^{-1}x+\tan^{-1}y$, find the Jacobian $\frac{\partial(f,g)}{\partial(x,y)}$. (5) CO3 BL3

1/7/2025 11:52:26 AM